

Question 1 30 marks in total

Q1: The following is an adjacency matrix for a graph G with vertices v_1, v_2, v_3, v_4, v_5 :

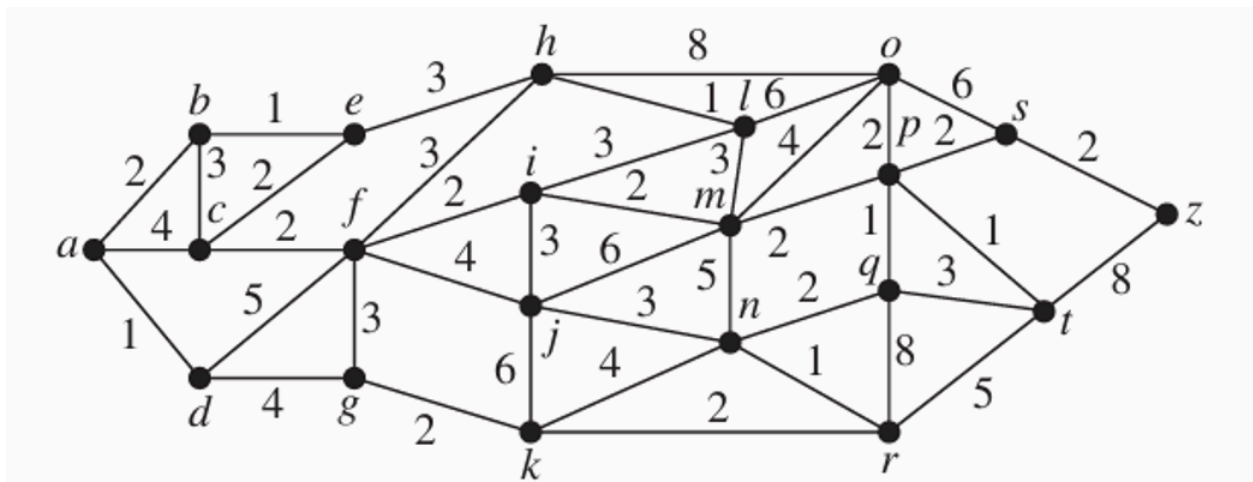
$$\begin{bmatrix} 0 & 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Answer the following questions by examining the matrix and its powers only, not by drawing the graph: (8 Marks)

- How many walks of length 2 are there from v_1 to v_4 ? 4
- How many walks of length 2 are there from v_3 to v_3 ? 1
- How many walks of length 3 are there from v_2 to v_5 ? 3
- How many walks of length 4 are there from v_5 to v_1 ? 7

Q2. Find the length of the shortest path from vertex "a" to vertex "z".

4 marks

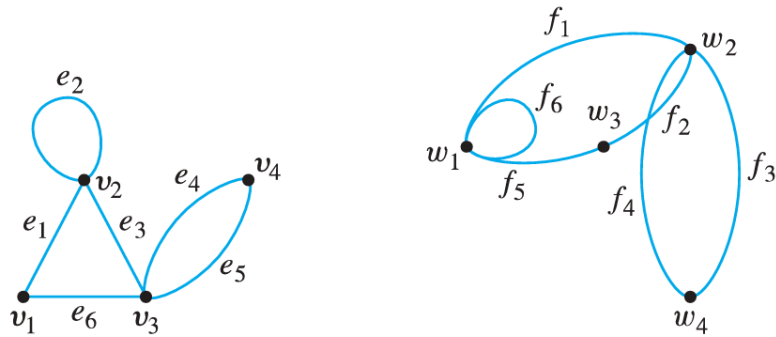


Answer: 16 is shortest length

path 1 : a-b-e-h-l-m-p-s-z ;

path 2: a-c-f-i-m-p-s-z

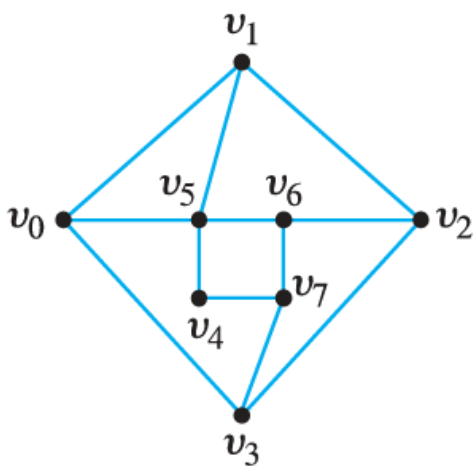
Q3: State either following graphs are isomorphic or not. If they are isomorphic, use arrows to show vertex correspondence. (4)



Answer:
 $v_1 \rightarrow w_3$
 $v_2 \rightarrow w_1$
 $v_3 \rightarrow w_2$
 $v_4 \rightarrow w_4$

v_1	w_1
v_2	w_2
v_3	w_3
v_4	w_4

Q4: Find if hamiltonian circuit exists, give reason if it doesn't exist. (4)



Yes it has Hamiltonian Circuit
Path : basically starting from any vertex and ending on the same is possible
$v_0-v_1-v_5-v_4-v_7-v_6-v_2-v_3-v_0$
$v_1-v_5-v_4-v_7-v_6-v_2-v_3-v_0-v_1$

Q5: Check if the following relations are Reflexive, Symmetric and Transitive. (10 MARKS)

1. $R_1 = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1)\}$
2. $R_2 = \{(1,1), (2,2), (3,3), (1,3), (3,2)\}$
3. $R_3 = \{(1,1), (2,2), (3,4), (4,3), (3,3), (4,4)\}$
4. $R_4 = \{(1,2), (2,3), (1,3), (4,4)\}$
5. $R_5 = \{(1,1), (2,2), (3,3), (4,4), (2,3), (3,2), (2,4), (4,2), (3,4), (4,3)\}$
6. $R_6 = \{(1,1), (2,2), (3,3)\}$
7. $R_7 = \{(1,4), (4,1), (2,3), (3,2), (1,1), (2,2), (3,3), (4,4)\}$
8. $R_8 = \{(1,2), (2,1), (2,3), (3,2)\}$
9. $R_9 = \{(1,1), (2,2), (3,3), (4,4), (1,4), (4,1)\}$
10. $R_{10} = \{(1,3), (2,4), (1,4)\}$

Important instruction : If someone has left the columns blank by only ticking under whatever property holds, it means the left column is considered a cross (property doesn't exist) in that case if somewhere property exists but the column is blank, marks are reduced.

Relations	Reflexive	Symmetric	Transitive
R1	yes	yes	yes
R2	yes	no	no
R3	yes	yes	yes
R4	no	no	yes
R5	yes	yes	yes
R6	yes	Vacuously yes	Vacuously yes
R7	yes	Yes	yes
R8	no	yes	no
R9	yes	yes	yes
R10	no	no	no